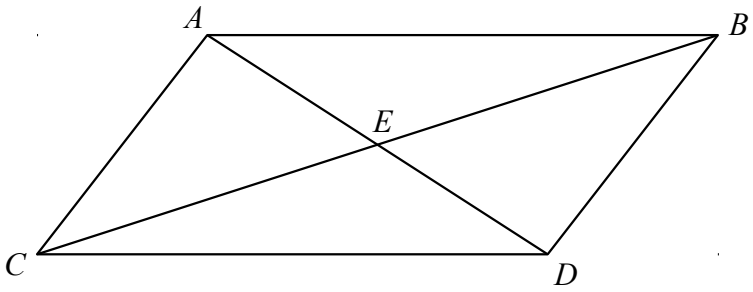
1 *ABCD* is a parallelogram



Prove that triangle *ABC* is congruent to triangle *BCD*.

(Total for Question 1 is 3 marks)

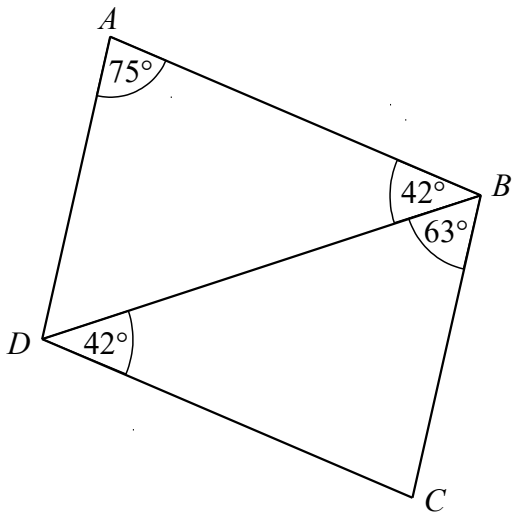
2 *ABCD* is a parallelogram
E is the point where the diagonals *AD* and *BC* meet.



Prove that triangle *ACE* is congruent to triangle *BDE*.

(Total for Question 2 is 3 marks)

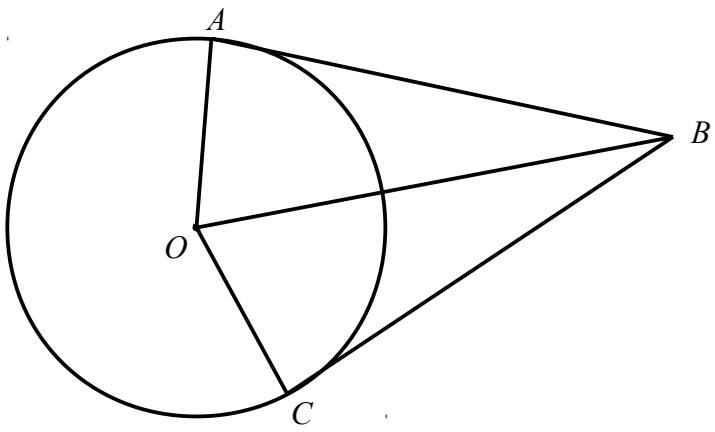
3 The diagram shows two triangles, *ABD* and *BCD*.



Prove that triangle *ABD* is congruent to triangle *BCD*.

(Total for Question 3 is 3 marks)

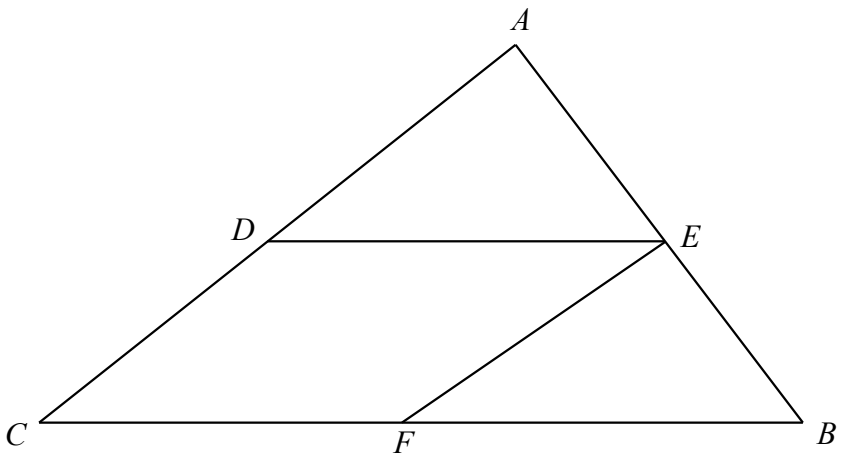
4 A and C are points on a circle, centre O.
AB and BC are tangents to the circle.



Prove that triangle ABO is congruent to triangle BCO.

(Total for Question 4 is 4 marks)

5 *ABC* is a triangle.

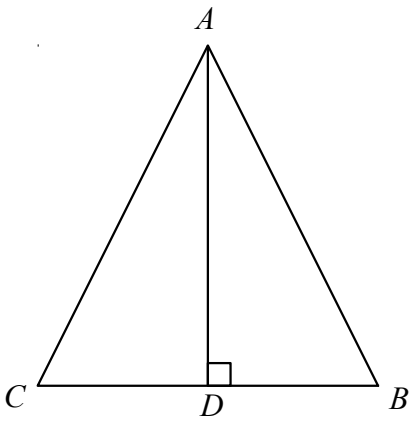


CDEF is a parallelogram such that:
D is the midpoint of *AC*
E is the midpoint of *AB*
F is the midpoint of *BC*

Prove that triangle *ADE* is congruent to triangle *BEF*.

(Total for Question 5 is 4 marks)

6 *ABC* is an equilateral triangle.

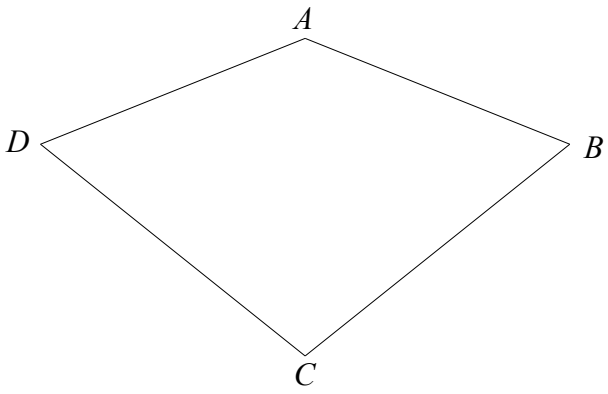


D lies on *BC*
AD is perpendicular to *BC*

Prove that angle *CAD* is equal to angle *BAD*.

(Total for Question 6 is 4 marks)

7 *ABCD* is a quadrilateral



$AB = AD$
$BC = CD$

Prove that angle ABC is equal to angle ADC.

(Total for Question 7 is 4 marks)